# Design from place.

Environment-aware camouflage design

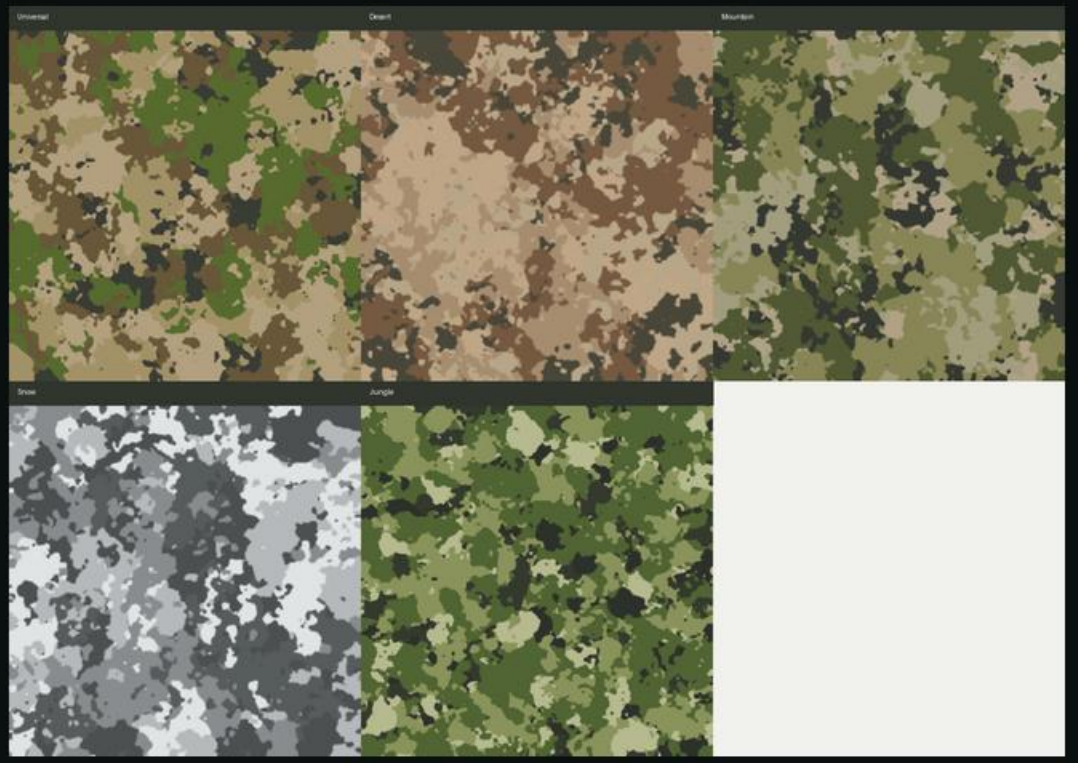


TerrainWeave / a visual research prototype

## Can environments guide camouflage design?

Reference photographs -> analysis -> pattern generation -> human and computer validation -> optimization

Current baseline patterns





# From pixels to palette.

A transparent, reproducible baseline

## INPUT

Representative landscape photos for forest, desert, mountain, snow, urban, jungle, taiga, and tundra.

->

## ANALYSIS

Dominant colors, luminance, local contrast, texture, and relevant terrain pixels.

->

## OUTPUT

A 500 x 500 finite-palette pattern with irregular connected regions.

Water and sky can be excluded or down-weighted when they are not relevant to the intended terrain.

# Make the pattern wearable.

A flat square is not enough

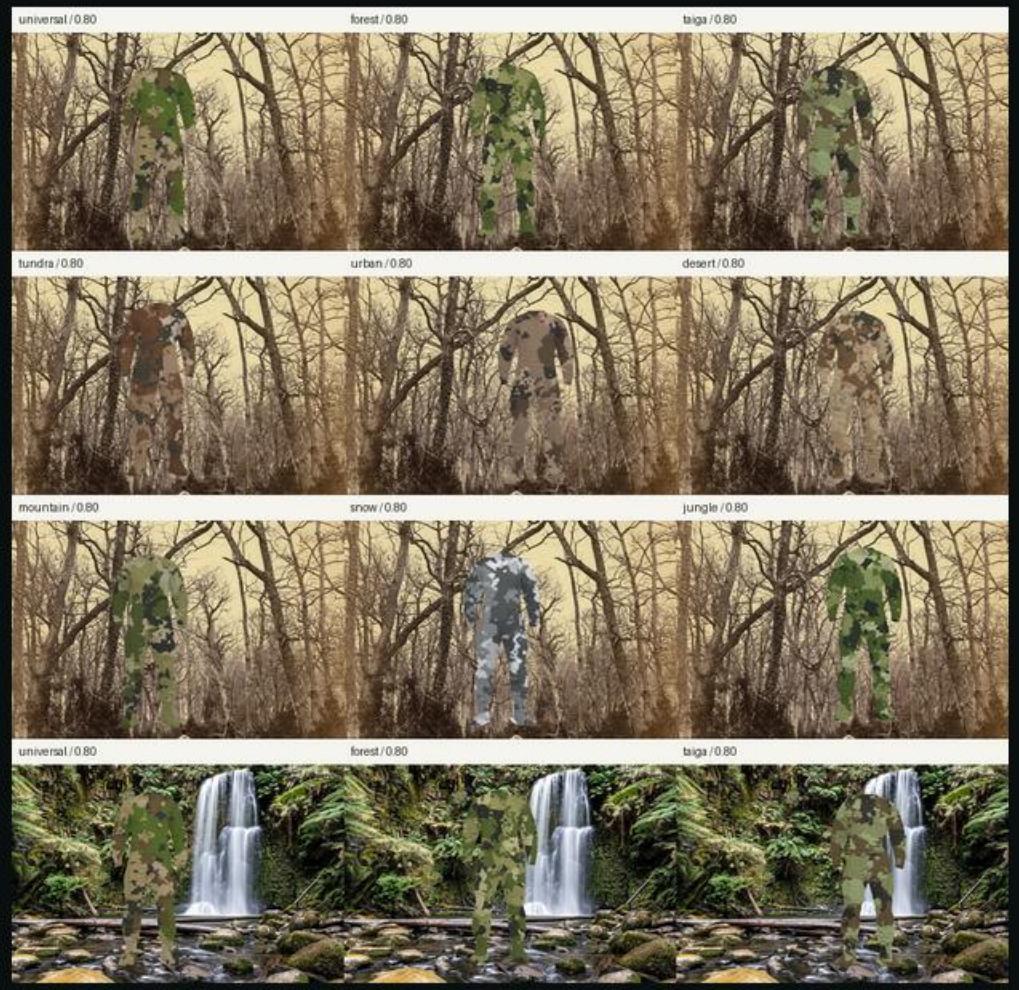


## WHAT WE CHECK

- overly large patches
- excessive dark colors
- stripes or repetitive edges
- behavior at different scales

# Does it disappear fast enough?

Recognition testing on real terrain




## HUMAN TEST

Short exposures and randomized locations measure how quickly observers notice the figure.

## COMPUTER METRICS

Color, luminance, edge, and visibility proxies compare candidate patterns.

## RULE

Keep the figure on terrain, not in obvious sky or water regions.



# Optimization is a loop.

Balance multiple scenes instead of fitting one photograph

A -> Generate candidates

B -> Score multiple scenes

C -> Compare human results

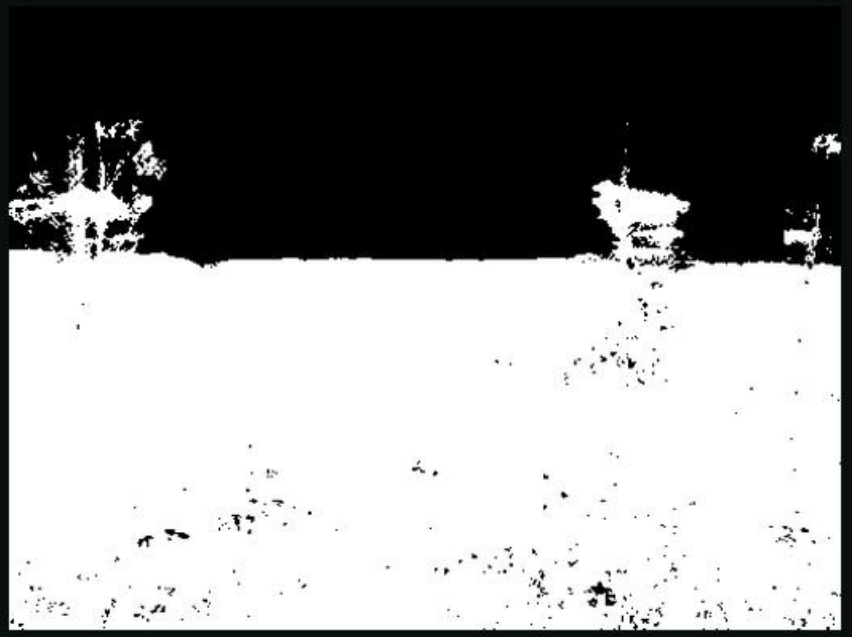
D Keep the best balanced pattern

photos -> initial pattern -> human/computer evaluation -> adjusted pattern -> repeat

Human CSV results are currently validation evidence. They are not yet used to train a neural network.

# Neural input. Procedural output.

SegFormer improves scene understanding; it does not generate the camouflage



## LANDSCAPE PHOTO

The pretrained model assigns semantic labels such as vegetation, terrain, sky, and water.

## RELEVANT PIXELS

The pipeline can focus color and texture extraction on the terrain that matters.

## LIMIT

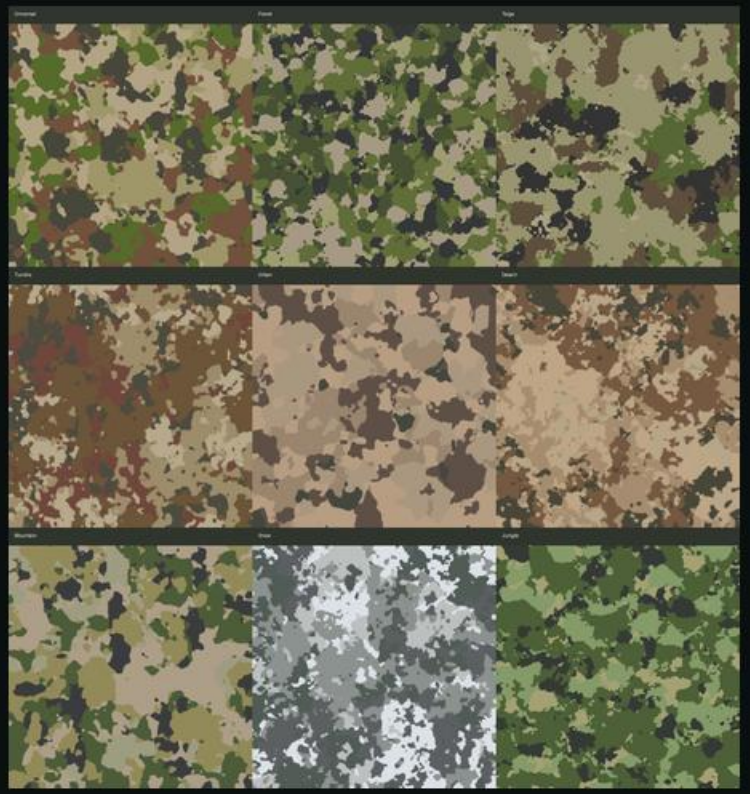
Segmentation improves the input analysis. It does not generate the camouflage or prove field performance.

# From idea to testable system.

A working research prototype with a reproducible workflow

- [OK] 9 environment profiles
- [OK] 124 reviewed evaluation images
- [OK] standard and weighted multi-objective generation
- [OK] optional neural segmentation
- [OK] uniform mockups and blinded human testing
- [OK] JSON configurations, tests, and reports

Optimized final patterns



Next milestone: learn a ranking model from a larger human-test dataset while keeping the procedural baseline.

# Three examples, one iteration.

Initial pattern -> optimized pattern -> measurable change

| | INITIAL | | OPTIMIZED | PROXY SCORE |
|---|---|---|---|---|
| UNIVERSAL |  | -> |  | 0.107 -> 0.094<br>change: +0.013 |
| DESERT |  | -> |  | 0.094 -> 0.094<br>change: +0.000 |
| FOREST |  | -> |  | 0.067 -> 0.060<br>change: +0.007 |

Lower proxy score is better within this controlled experiment. The full nine-environment comparison remains available in output/results_comparison/.